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SEAMLESS CIRCULAR KNIT FABRIC, A METHOD OF MAKING SAME AND A FORMING MACHINE USING THIS METHOD, AS WELL AS ARTICLES COMPRISING STRAPS OR BANDS MADE THEREFROM

Abstract

A seamless circular knit fabric (**10**) comprising an elastic yarn is provided. The fabric (**10**) has a continuous circular knit through the length thereof. The circular knit fabric (**10**) further includes a cured cushion filling (**12**) inside the circular knit fabric (**10**) and bonded thereto. The method of making same comprises knitting a yarn to produce a continuous length of circular knit fabric (**10**), and injecting a curable cushion filling in a pre-cured state inside the circular knit fabric (**10**). This method uses a circular knit fabric forming machine assembly comprising a circular knitting machine and an injector.

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Background/Summary

[0001] This invention relates to a seamless circular knit fabric, a bra comprising a shoulder strap made from such seamless circular knit fabric, a method of forming a seamless circular knit fabric and a seamless circular knit fabric forming machine assembly.
[0002] Shoulder straps of a bra are typically formed from strips of elastic material which have uncomfortable and conspicuous edge binding material. These materials are relatively hard on a wearer's skin. Traditional shoulder straps tend to dig into the wearer's skin especially as bras are worn by the wearer for the majority of the day. It is also known to use similar strips of material for ear loops which may be used for positioning a face mask onto a wearer's face. Again, the relatively hard ear loop elastic material has a tendency to dig into the wearer's face which is not only uncomfortable but may also leave marks on the skin. The same can be said for the underband of a bra (the part the extends around the back of the wearer) and stretchable waistbands, e.g. for briefs, pants, shorts or trousers/leggings.
[0003] According to a first aspect of the invention there is provided a seamless circular knit fabric comprising an elastic yarn, the fabric having a continuous circular knit through the length thereof, the circular knit fabric further including a cured cushion filling inside the circular knit fabric and bonded thereto.
[0004] A seamless circular knit fabric in the context of the invention is understood to be a seamless tubular fabric. This is different to joining together two pieces of fabric or two edges of the same fabric to produce a tubular fabric, which would not be seamless because of the joint.
[0005] The fabric having a continuous circular knit through its length means that the circular knit pattern flows continuously throughout the length of the fabric. The fabric can be cut to any size, depending on the desired application of the fabric, to create a discrete length of circular knit fabric. Therefore, a user is not restricted by a pre-set length of tubular fabric which circular knitting machines typically produce (e.g. a circular knit sock machine). Moreover, such typical circular knitting machines tend to release a finished single piece of fabric at a pre-set length (e.g. a sock which has its heel and toe knitted), such that the single piece of fabric is not considered to have a continuous circular knit through the length thereof.
[0006] Meanwhile, the seamless circular knit fabric having a cured cushion filling provides additional comfort to the circular knit fabric which may be used as or as part of a wearable garment or accessory. The cured cushion filling being inside the continuous fabric and bonded thereto means that the filling is integral to the fabric and thus provides robust and consistent comfort throughout the length of the fabric. In particular, the filling is added during the manufacturing

process of the circular knit fabric which is why is it present in the continuous length of circular knit fabric and bonded thereto (as opposed to it being added after as a separate component to the fabric). The cushion filling being “cured” is understood to mean that it is made from a material which was previously in a non-cured (e.g. soft or liquid) state prior to it curing. As explained in more detail later on, this means that the cushion filling can be added to the circular knit fabric and cured during the knitting process.

[0007] Preferably, the cured cushion filling is a thermoplastic or thermosetting polymer. Optionally, the cured cushion filling is a two-part silicone gel.

[0008] Any such arrangement allows for the cushion filling to be in a pre-cured state during manufacture of the circular knit fabric which allows it to flow into and fill the hollow cavity of the circular knit fabric as it is being made. It also permits the shape of the fabric (with the pre-cured cushion filling therein) to be moulded to a particular shape, as described in more detail later on.

[0009] The seamless circular knit fabric may have a cut end to define a discrete length of circular knit fabric.

[0010] By the fabric having a cut end it is understood that the fabric as a whole is cut, i.e. the yarn through the circumference of the fabric is cut, so as to create a discrete length of continuous circular knit fabric which can be any desired length. This differs from a single piece of fabric that is released from a machine by cutting the yarn fed into the machine. Such single pieces of fabric would have the knitting finished by the machine at the pre-set length before being released, such that the single piece of fabric is not considered to have a cut end.

[0011] Optionally the seamless circular knit fabric has more than one filled portion, the filled portion being filled with the cured cushion filling, wherein each filled portion is separated by a non-filled portion, the non-filled portion being absent of the cured cushion filling.

[0012] Such an arrangement of neighbouring filled and non-filled portions of the circular knit fabric allows for ease of cutting the continuous length of circular knit fabric at the non-filled portions to form discrete lengths of fabric. Moreover, such cutting would result in the discrete lengths of fabric having non-filled ends which makes it easier to attach the circular knit fabric to another fabric or garment.

[0013] Preferably, the seamless circular knit fabric may have a minimum width of 2 mm, optionally 6 mm, optionally 8 mm. The seamless circular knit fabric may have a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0014] Such minimum widths permit the circular knit fabric to be suitable for use as an ear loop (e.g. minimum of 2 mm) or a bra shoulder strap (e.g. minimum of 6 or 8 mm).

[0015] According to a second aspect of the invention, there is provided a bra comprising a body portion for positioning around a torso of a wearer, and a shoulder strap secured at each end to the body portion, the shoulder strap being made from the seamless circular knit fabric as defined hereinabove.

[0016] According to a third aspect of the invention, there is provided a bra comprising an underband portion for positioning around a torso of a wearer, the underband portion being made from the seamless circular knit fabric as defined hereinabove.

[0017] According to a fourth aspect of the invention there is provided a bottom garment comprising a stretchable waistband for positioning around a waist of a wearer and a bottom portion extending from the waistband, the waistband being made from the seamless circular knit fabric as defined hereinabove.

[0018] It will be understood that the bottom portion may extend from the waistband down the full length of the wearer's leg (e.g. trousers or leggings) or only a portion of the length of the wearer's leg (e.g. shorts, boxer shorts, skirt, cropped trousers or leggings) or only as far as the wearer's buttocks (e.g. pants, briefs).

[0019] According to a fifth aspect of the invention there is provided a face mask comprising a face covering portion for positioning over a wearer's nose and mouth, and a head strap secured at each

end to the face covering portion, the head strap being made from the seamless circular knit fabric as defined hereinabove.
[0020] It will be understood that the “head strap” may be a strap that goes all the way around the wearer's head or may loop around the wearer's ear (i.e. two ear loops).
[0021] Using the seamless circular knit fabric as described hereinabove as a bra shoulder strap, bra underband, waistband or a face mask head strap provides optimal comfort to the wearer firstly because there are no joints in the strap/band that may irritate or dig into the wearer's skin and secondly because the cushion filler provides additional softness and comfort to the wearer.
[0022] According to a sixth aspect of the invention there is provided a method of forming a seamless circular knit fabric comprising: [0023] feeding an elastic yarn into a circular knit machine head, [0024] knitting the yarn by bending the yarn into loops and intermeshing yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric, [0025] injecting a curable cushion filling in a pre-cured state inside the circular knit fabric as the fabric is being knitted.
[0026] The advantages set out above in relation to the features of the first aspect of the invention apply mutatis mutandis to the fourth aspect of the invention.
[0027] It will be understood that injecting the curable cushion filling in a pre-cured state inside the circular knit fabric as the fabric is being knitted means that the pre-cured filling is injected into the hollow cavity formed by the circular knit fabric.
[0028] Preferably, the step of injecting a curable cushion filling includes injecting a thermoplastic or thermosetting polymer.
[0029] Optionally the step of injecting a curable cushion filling includes injecting a two-part silicone gel.
[0030] In an embodiment, the method includes curing the cushion filling as the continuous length of circular knit fabric is being made.
[0031] By curing the cushion filling as the continuous length of circular knit fabric is being made, the cured cushion filling is formed during the same knitting process as that forms the circular knitted fabric, thus reducing the processing steps required. Moreover, the curing step results in a cured cushion filling that forms part of (i.e. bonded to) the circular knit fabric, thus providing a continuous and consistent cushioning throughout the length of the circular knit fabric.
[0032] Optionally the method includes sequentially filling and ceasing to fill the circular knit fabric with the curable cushion filling in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric.
[0033] The method may include passing the continuous length of circular knit fabric with curable cushion filling through an adjustment mechanism to adjust the shape of the fabric.
[0034] Such a step makes use of the mouldable nature of the pre-cured cushion filling to produce a circular knit fabric with a desired shape.
[0035] The method may include feeding the yarn from a yarn platform upwards towards the circular knit machine head, and producing the continuous length of circular knit fabric in a space created between the circular knit head and the yarn platform.
[0036] Such an arrangement provides space for the circular knit fabric to be produced in a continuous manner without restricting the length of the fabric being produced. Moreover, creating a space between the yarn platform and the circular knit machine head allows for additional processing steps to readily take place during the knitting process because such additional mechanisms (e.g. an adjustment mechanism) can operate in that space between the yarn platform and the circular knit machine head. In addition to the foregoing, it allows the injection of the pre-cured cushion filling to happen directly at the machine head without obstruction from other parts of the machinery. This is important to prevent premature curing of the cushion filling which may otherwise occur of the cushion filling has a further distance to travel to the machine head.
[0037] The method may include pulling the continuous length of circular knit fabric in the direction

that the circular knit fabric is being produced.

[0038] Such a pulling step (e.g. performed by a takedown mechanism) helps to dictate the courses per inch (CPI) or the fabric.

[0039] Preferably, the method further includes cutting the continuous length of circular knit fabric to discrete lengths of circular knit fabric.

[0040] Cutting the continuous length of circular knit fabric permits the creation discrete lengths of continuous circular knit fabric which can be any desired length.

[0041] The method may include producing a continuous length of circular knit fabric having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm. The method may include producing a continuous length of circular knit fabric having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0042] According to a seventh aspect of the invention there is provided a seamless circular knit fabric forming machine assembly comprising: [0043] a circular knit machine head configured to bend elastic yarn into loops and intermesh the yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric, [0044] a yarn feeder configured to feed yarn into the circular knit machine head, [0045] an injector configured to inject a curable cushion filling in a pre-cured state inside the circular knit fabric as it is being knitted.

[0046] The advantages set out above in relation to the features of the first and fourth aspects of the invention apply mutatis mutandis to the fifth aspect of the invention.

[0047] The injector may be configured to inject a thermoplastic or thermosetting polymer.

[0048] The injector may be configured to inject a two-part silicone gel.

[0049] Optionally the injector is configured to sequentially fill and cease to fill the circular knit fabric with the curable cushion filling in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric.

[0050] Optionally, the machine assembly includes a curing mechanism configured to apply a curing effect to the continuous circular knit fabric to achieve curing of the pre-cured cushion filling.

[0051] Preferably, the machine assembly includes an adjustment mechanism configured to receive the continuous circular knit fabric with pre-cured cushion filling and to adjust the shape of the fabric as a result of the fabric passing through the adjustment mechanism.

[0052] The machine assembly may include yarn platform wherefrom the yarn is fed upwards towards the circular knit machine head, wherein the continuous circular knit fabric is produced in a space created between the circular knit head and the yarn platform.

[0053] The machine assembly may include a takedown mechanism configured to pull the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.

[0054] Optionally the machine assembly includes a cutting mechanism configured to cut the continuous length of circular knit fabric into discrete lengths of fabric.

[0055] The circular knit machine head may be configured to produce a continuous length of circular knit fabric having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm.

[0056] The circular knit machine head may be configured to produce a continuous length of circular knit fabric having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

[0057] A preferred embodiment of the invention will now be described, by way of a non-limiting example, with reference to the accompanying drawings in which:

Description

[0058] FIG. **1***a* shows a seamless circular knit fabric according to a first embodiment of the invention;

[0059] FIG. **1***b* shows the seamless circular knit fabric according to a second embodiment of the

invention;
[0060] FIGS. **2***a* and **2***b* show a seamless circular knit fabric forming machine assembly according to a third embodiment of the invention;
[0061] FIG. **3***a* shows the needle bed of the machine head from the machine assembly shown in FIGS. **2***a* and **2***b;*
[0062] FIG. **3***b* shows the direction of operation of the needle bed shown in FIG. **3***a;*
[0063] FIG. **4** shows an injector portion of the machine assembly shown in FIGS. **2***a* and **2***b*; and
[0064] FIG. **5** shows an adjustment and curing portions of the machine assembly shown in FIGS. **2***a* and **2***b.*
[0065] A seamless circular (or tubular, as it may also be referred to) knit fabric according to an embodiment of the invention is shown in FIGS. **1***a* and **1***b* and is designated by reference numeral **10**.
[0066] The circular knit fabric **10** has a degree of stretch due to the inclusion of an elastic yarn. The circular knit fabric **10** has a continuous circular knit through the length thereof, which is discussed in more detail later.
[0067] The circular knit fabric **10** further includes a cured cushion filling **12** inside the circular knit fabric **10**. That is to say that the cured cushion filling **12** fills what would be the hollow cavity **14** of the circular knit fabric **10**. The cured cushion filling **12** is bonded to the circular knit fabric **10**.
[0068] In the embodiment shown, the cured cushion filling is a thermoplastic elastomer (TPE) and more specifically may be thermoplastic polyurethane (TPU).
[0069] In other embodiments, the cured cushion filling **12** may instead by any suitable thermoplastic or thermosetting polymer. The cured cushion may be a two-part silicone gel. The silicone gel may be of a type known as RTV 2K GEL TP 3841™ or “SILOPREN” gel TP 3904 as sold by GE Bayer Silicones GmbH & Co. KG. The cured cushion filling may instead be a curable foam or a polymer gel. The cured cushion filling may instead be a bio-based polymer, biodegradable synthetic polymer, or naturally occurring biodegradable/non-degradable polymer.
[0070] The circular knit fabric **10** has a cut end **16** which defines a discrete length of circular knit fabric. The circular knit fabric **10** leaves the knitting machine as a continuous length of circular knit fabric **10**, which can then be cut into desired lengths. This is opposed to the circular knit fabric being finished as a pre-set length single piece of fabric.
[0071] In this embodiment, the cured cushion filling **12** fills the length of circular knit fabric **10** in a continuous manner such that even when it is cut, the cured cushion filling **12** fills right up to the cut end **16**.
[0072] In the embodiment shown, the width of the circular knit fabric **10** is approximately 8 mm. The width may instead be as narrow as 2 mm (e.g. ear loop) or may be as wide as 50 mm (e.g. waistband). The continuous length of circular knit fabric **10** that is produced may vary but may be at least 40 mm. The output table below indicates exemplary widths of the circular knit fabric **10** and an intended end use of a fabric having such a width. The table also outlines an example of the output per hour and speed of the circular knit machine that may be used (which is discussed in more detail later on).
TABLE-US-00001 Output Table 1 Circular knit Output per Speed of fabric width hour machine End use 2 mm-6 mm 1000 RPM Ear loop 8 mm-18 mm Ave. 9M Per one 700 RPM Bra Strap machine head 20 mm-30 mm Ave. 9M Per one 600 RPM Bra Strap machine head
[0073] The finished circular knit fabric width may have the following tolerances: [0074] 2 mm-6 mm accurate [0075] 8 mm to 18 mm: +/−0.5 mm [0076] 20 mm to 30 mm: +/−1 mm
[0077] The circular knit fabric may be formed from a single elastic yarn. The elastic yarn may be Nylon. For example, the elastic yarn may be 2/78/24/0 Nylon 66 (“2/78/24/0” being a common way to represent characteristics of a yarn, i.e. representing 2 ply, 78 denier, 24 filaments, 0 twists per meter (TPM)).
[0078] The circular knit fabric may be formed from more than one yarn. Both or all yarns may be

the same type of yarn or they may instead be different yarns. The yarns may have different properties. For example, the yarn may be any single cover yarn.

[0079] A circular knit fabric **10′** according to a second embodiment of the invention is shown in FIG. **1***b*. The circular knit fabric **10′** of the second embodiment shared identical features to the circular knit fabric **10** of the first embodiment and like features are indicated by the same reference numerals.

[0080] As shown in FIG. **1***b*, the circular knit fabric **10′** in this embodiment includes first and second filled portions **18***a*, **18***b* which are filled with the cured cushion filling **12**. The first and second filled portions **18***a*, **18***b* are separated by a non-filled portion **20** which is absent of any cured cushion filling **12**.

[0081] Although only two filled portions **18***a*, **18***b* are shown in FIG. **1***b*, it will be understood that there may be any number of filled portions **18***a* **18***b* separated by non-filled portions **20**.

[0082] The non-filled portion **20** is considerably shorter in length to the filled portions **18***a* **18***b*. In other embodiments, the lengths of the filled and non-filled portions **18***a*, **18***b*, **20** may be similar or the same as each other. In further embodiments, the lengths of each portion **18***a*, **18***b*, **20** may differ (i.e. the first filled portion **18***a* may have a different length to the second filled portion **18***b*).

[0083] Moreover, each of the first and second filled portions **18***a*, **18***b* may be filled with different types of cured cushion filling **12**.

[0084] The non-filled portion **20** presents a section which can be easily cut (i.e. at line **22**) to form discrete lengths of circular knit fabric **10′**, each of which having a non-filled cut end.

[0085] A seamless circular knit fabric forming machine assembly is shown in FIGS. **2***a* to **5** and is designated by reference numeral **100**.

[0086] The machine assembly **100** has a circular knit machine head **102** secured to a knit head platform **104**. The machine assembly **100** also includes a yarn feeder **106** configured to feed yarn **107** into the circular knit head machine **102**.

[0087] The machine assembly **100** further includes a yarn platform **108** from which the yarn **107** is supplied. In particular, the yarn platform **108** includes bobbin holders **110** (in the form of pegs **110** in this example) onto which yarn bobbins **112** locate in use. In the example shown, there are four pegs **110** and therefore four yarn bobbins **112** are positioned on the yarn platform **108**. The yarn bobbins **112** are stood in an upright manner relative to the yarn platform **108**. As can be seen, the yarn pegs **110** are positioned around the centre of the yarn platform **108**.

[0088] The knit head platform **104** and the yarn platform **108** are spaced from one another such that a gap **114** is created between the circular knit machine head **102** and the yarn platform **108**. Moreover, the knit head platform **104** is positioned higher than the yarn platform **108**. In other words, there is a space **114** between where the yarn **107** is being supplied and where the yarn **107** is being fed into the machine, and the yarn **107** is being fed to the feeder **106** in an upwards manner.

[0089] In the embodiment shown, the knit head platform **104** is positioned directly above the yarn platform **108**. Moreover, the knit head platform **104** and the yarn platform **108** are centrally aligned with one another. Both platforms **108**, **104** are circular platforms which share a central axis C. In other embodiments, the platforms **108**, **104** may take any shape and/or may be out of line with one another.

[0090] The space **114**, i.e. the distance between the yarn platform **108** and the knit head platform **104** may be around 40 cm.

[0091] The machine assembly **100** includes a support member **116** which secures the platforms **104**, **108** to one another and spatially positions them relative to one another. In the embodiment shown, there are four support members **116** that are equally spaced around the centre of each platform **104**, **108**. The support member **116** may take any suitable form.

[0092] The machine assembly **100** includes yarn guides **118** which guide the yarn **107** from the yarn platform **108** towards the yarn feeder **106**. The yarn guides **116** help to maintain a desired tension in the yarn **107**.

[0093] In the embodiment shown, there are four yarn guides **118** which are secured to the knit head platform **104**. Each yarn guide **116** is equally spaced from one another around the centre of the knit head platform **104**. Each yarn guide **118** positioned in line with a peg **110** and receives yarn **107** from the yarn bobbin **112** which is situated on the peg **110** directly below the respective yarn guide **118**. There is a single yarn feeder **106** and the yarn guides **118** feed yarn **107** into the single yarn feeder **106**.
[0094] In the embodiment shown, the yarn **107** from two of the yarn bobbins **112** is fed directly from the yarn bobbin **112** through the respective yarn guide **118** to the yarn feeder **106**. Meanwhile the yarn **107** from the remaining two of the yarn bobbins **112** is fed from the yarn bobbin **112** through the respective yarn guide **118** and then via a neighbouring yarn guide **118** to the yarn feeder **106**. This is shown more clearly in FIG. **2***b* in which the direct yarn path Y.sub.D from two of the yarn bobbins **112***a* is shown and the indirect yarn path Y.sub.I of the remaining two yarn bobbins **112***b* is shown.
[0095] Turning to FIGS. **3***a* and **3***b*, the circular knit machine head **102** is configured to bend the yarn **107** into loops **120** and intermesh the yarn loops **120** in a horizontal direction D in a seamless manner so as to produce a continuous length of circular knit fabric.
[0096] The circular knit machine head **102** includes a needle bed **122** which holds a plurality of needles **124** in a cylindrical manner (as shown in FIG. **3***a*-only two needles shown for clarity purposes). The needles **124** have a hook **125** at one end. The needles **124** are held at specific distances relative to one another, which is determined by the needle bed **122**. The needle bed **122** also guides the needles **124** during the knitting process. To achieve this control, the needle bed **122** includes needle slots **126** (also referred to as needle tracks) in which a respective needle **124** is located and the slots **126** act to guide each needle **124** during knitting.
[0097] As can be seen in FIG. **3***b*, the needles **124** move up and down in turn in a horizontal (when looking directly at the machine head **102**) direction of travel D, with each needle **124** moving up to catch (via its hook **125**) the yarn **107** being fed into the circular knit machine head **102** and then pulling that yarn **107** down and through a yarn loop **120** already created in the fabric.
[0098] The number of slots **126** in the circular knit machine head **102** determines the knitting gauge. The gauge may be 12 to 16.
[0099] The diameter of the circular knit machine head **102** influences the width of the circular knit fabric **10** that is produced. The type of yarn **107** that is used is also a factor in the final width of the fabric **10**.
[0100] Turning to FIG. **4**, the machine assembly **100** further includes an injector **128** which is configured to inject a curable cushion filling **129** in a pre-cured state inside the circular knit fabric **10** as it is being knitted.
[0101] The injector **128** includes an extrusion machine **130** and a nozzle **132**. In this embodiment, the extrusion machine **130** extrudes TPE and the nozzle **132** is an inkjet nozzle which inserts TPE in a pre-cured state into the circular knit fabric **10**.
[0102] As can be seen, the nozzle **132** points into the hollow cavity **14** of the circular knit fabric **10** at the end where the fabric **10** is being formed.
[0103] In some embodiments, the injector **128** may be configured to sequentially fill and cease to fill the circular knit fabric with the curable cushion filling **129** in a pre-cured state to form neighbouring filled and non-filled portions of the circular knit fabric. Such filled and non-filled portions are described in more detail above in relation to FIG. **1***b*.
[0104] Turning to FIG. **5**, the machine assembly **100** further includes an adjustment mechanism **134**. The adjustment mechanism **134** shown in this embodiment is a tapered cavity **136** which receives, at the wider end **138** thereof, the continuous circular knit fabric **10** with the pre-cured cushion filling inside. The fabric **10** passes through the narrower end **140** of the tapered cavity **136** which moulds the shape of the pre-cured cushion filling and therefore the shape of the circular knit fabric **10**. For example, the tapered cavity **136** may be an oblong shape at its narrow end **140** and

therefore squash the fabric **10** to a flat, thin shape. Other shapes that the adjustment mechanism **134** may produce include oval, circular, rectangular.

[0105] As also shown in FIG. **5**, the machine assembly **100** further includes a curing mechanism **142**. The curing mechanism **142** is located directly below the adjustment mechanism **134** so as to cure the cushion filling after it has been moulded to a desired shape by the adjustment mechanism **142**, therefore setting the desired shape of the circular knit fabric **10**.

[0106] In this embodiment, the curing mechanism **142** is an air cooler **144** which blows cool air over the circular knit fabric **10** to cure the TPE cushion filling. In other embodiments, the curing can be done by cool or hot air, water or by adding another material which initiates curing. Moreover, the curing mechanism **142** may be a combination of such curing methods.

[0107] Although not shown in this embodiment, the machine assembly **100** may also include a takedown mechanism which is configured to pull the circular knit fabric **10** down in the direction of travel (i.e. in the direction it is being produced). The machine assembly **100** may also include a cutting mechanism to cut the continuous length of circular knit fabric to desired discrete lengths.

[0108] Exemplary raw materials and machine settings to produce a circular knit fabric **10** of the invention are outlined below.

Example 1

[0109] Intended use: Bra strap

Raw Material Information:

TABLE-US-00002 Raw materials 1 Yarn info Yarn code No of Yarn Yarn weight Yarn 2/78/24/0 Nylon 66 semi 3784 2 2.6 g 1 dull textured set air mingle Yarn Single cover yarn 2 2.1 g 2

Machine Settings

[0110] Loom speed: 700 RPM [0111] Loom picks: 3.6 [0112] Unfinished width: 11 mm [0113] Number of tape: 10 [0114] Unfinished stretch: 90 at 3.5 Kg [0115] Finished picks (courses): 33 [0116] Finished weight (1 M): 5.1 g [0117] Machine head diameter: 17 mm [0118] Number of needles: 34 [0119] Needle gauge: 16 [0120] Noodle size: 16

Claims

1. A seamless circular knit fabric comprising an elastic yarn, the fabric having a continuous circular knit through a length thereof, the circular knit fabric further including a cured cushion filling that fills an inside of a cavity of the circular knit fabric and bonded thereto.

2. The seamless circular knit fabric according to claim 1 wherein the cured cushion filling is a thermoplastic or thermosetting polymer.

3. The seamless circular knit fabric according to claim 1 wherein the cured cushion filling is a two-part silicone gel.

4. The seamless circular knit fabric according to claim 1 having a cut end to define a discrete length of circular knit fabric.

5. The seamless circular knit fabric according to claim 1 having more than one filled portion, the filled portion being filled with the cured cushion filling, wherein each filled portion is separated by a non-filled portion, the non-filled portion being absent of the cured cushion filling.

6. The seamless circular knit fabric according to claim 1 having a minimum width of 2 mm, optionally 6 mm, optionally 8 mm.

7. The seamless circular knit fabric according to claim 1 having a maximum width of 20 mm, optionally 30 mm, optionally 50 mm.

8. The seamless circular knit fabric according to claim 1, wherein the fabric forms at least one of: a shoulder strap secured at each end to a body portion, the body portion for positioning around a torso of a wearer in use; an underband portion of a bra, the underband portion for positioning around a torso of a the wearer in use; a stretchable waistband of a bottom garment, the stretchable waistband for positioning around a waist of the wearer; or a head strap secured at each end to a face

covering portion of a face mask.
9. (canceled)
10. (canceled)
11. (canceled)
12. The method according to claim **34** wherein the step of inserting the curable cushion filling includes: injecting the curable cushion filling in a pre-cured state inside the cavity of the circular knit fabric such that the cavity is filled with the curable cushion filling, wherein the injecting is being carried out as the fabric is being knitted.
13. (canceled)
14. (canceled)
15. The method according to claim **34**, including curing the cushion filling as the continuous length of circular knit fabric is being made.
16. (canceled)
17. The method according to claim **34** including passing the continuous length of circular knit fabric with the curable cushion filling through an adjustment mechanism to adjust a shape of the fabric.
18. The method according to claim **34** including feeding the yarn from a yarn platform upwards towards the circular knit machine head, and producing the continuous length of circular knit fabric in a space created between the circular knit head and the yarn platform.
19. The method according to claim **34** including pulling the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.
20. The method according to claim **34** further including cutting the continuous length of circular knit fabric to discrete lengths of circular knit fabric.
21. (canceled)
22. (canceled)
23. The seamless circular knit fabric forming machine assembly according to claim **35** wherein the inserter includes an injector configured to inject the curable cushion filling in a pre-cured state inside the cavity of the circular knit fabric so as to fill the cavity with the curable cushion filling as the circular knit fabric is being knitted.
24. (canceled)
25. (canceled)
26. (canceled)
27. (canceled)
28. The seamless circular knit fabric forming machine assembly according to claim **35**, further including an adjustment mechanism configured to receive the continuous circular knit fabric with the pre-cured cushion filling and to adjust a shape of the fabric as a result of the fabric passing through the adjustment mechanism.
29. The seamless circular knit fabric forming machine assembly according to claim **35**, including a yarn platform wherefrom the yarn is fed upwards towards the circular knit machine head, wherein the continuous circular knit fabric is produced in a space created between the circular knit head and the yarn platform.
30. The seamless circular knit fabric forming machine assembly according to claim **35**, further including a takedown mechanism configured to pull the continuous length of circular knit fabric in the direction that the circular knit fabric is being produced.
31. (canceled)
32. (canceled)
33. (canceled)
34. A method of forming a seamless circular knit fabric comprising: feeding an elastic yarn into a circular knit machine head; knitting the yarn by bending the yarn into loops and intermeshing yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit

fabric; inserting a curable cushion filling inside a cavity of the circular knit fabric such that the cavity is filled with the curable cushion filling, wherein the inserting is being carried out as the fabric is being knitted.

35. A seamless circular knit fabric forming machine assembly comprising: a circular knit machine head configured to bend elastic yarn into loops and intermesh the yarn loops in a horizontal direction in a seamless manner to produce a continuous length of circular knit fabric; a yarn feeder configured to feed yarn into the circular knit machine head; and an inserter configured to insert a curable cushion filling inside a cavity of the circular knit fabric so as to fill the cavity with the curable cushion filling as the circular knit fabric is being knitted.